CAPSTONE PROJECT SUBMISSION TEMPLATE

Template 7: Project Documentation

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 7: Project Documentation**
Student Name:	Arnaov Garg	**Email:**	ag5088@srmist.edu.in
Institution:	SRM Institute of Science and Technology (SRMIST)	**Track / Cohort:**	Snowflake Data Engineering (Snowflake Data Engineering_EL_2027_Cohort 1)
GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. Codebase Architecture & GitHub Repository Structure

Official GitHub Repository: https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics
The repository is organized following professional software engineering and data engineering standards:

File / Path	Repository Documentation Deliverable & Purpose
README.md	Executive platform overview, architecture diagrams, technology stack, metric scorecard, and step-by-step local setup guide.
docs/PROJECT_REPORT.md	Comprehensive 224-line end-to-end capstone report detailing all 8 epics, business findings, and SQL implementation.
docs/ARCHITECTURE.md	Data dictionary, Medallion layer specifications, schema ERD, primary/foreign key mappings, and lineage flows.
docs/KANBAN_TASK_MAPPING.md	Complete mapping guide for all 8 Epics and 20 Tasks conforming to the SmartBridge MySkillWallet portal.
sql/01_data_architecture_setup.sql	DDL provisioning warehouse, database, schemas (RAW, STAGING, CURATED), stage, and file formats.
sql/02_data_ingestion_raw.sql	DDL for 5 raw tables and COPY INTO ingestion commands with automated validation queries.
sql/03_data_preparation_staging.sql	Staging ETL performing type casting, string hygiene, and financial outcome classifications.
sql/04_data_modelling_curated.sql	Curated Star Schema DDL, MD5 surrogate keys, dimensions, facts, and persona data marts.
sql/05_business_analytics_kpis.sql	Production SQL queries computing platform KPIs, GMV trends, cuisine rankings, and rating elasticity.
sql/06_performance_testing.sql	Partition pruning benchmarks, query history profiling, cache hit verifications, and column audits.
streamlit_app/app.py	5-tab interactive Streamlit web dashboard powered by Plotly Express and dual-mode database connector.
run_pipeline_local.py	Automated Python driver script to initialize local database, load datasets, and run validation tests.

2. Step-by-Step Execution & Deployment Guide

- **Step 1 — Clone Repository:** git clone https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics.git
- **Step 2 — Install Dependencies:** pip install -r streamlit_app/requirements.txt
- **Step 3 — Run Pipeline:** Execute python run_pipeline_local.py to verify data generation and table loading.
- **Step 4 — Execute Snowflake Scripts:** Run SQL scripts 01_ through 06_ in the Snowflake Snowsight console.
- **Step 5 — Launch Dashboard:** Run streamlit run streamlit_app/app.py to launch the interactive UI in your browser.